An algorithmic answer for the MSS-type theorem in *Restricted invertibility revisited*

Literature-status packet for arXiv:1601.00948

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Source question

Naor–Youssef ask in the introduction of *Restricted invertibility revisited* (arXiv:1601.00948, source PDF page 8) whether their Theorems 6, 7, 9, and 11 can be made algorithmic. For an $n \times m$ matrix A of rank $R > k$, their Theorem 11 gives a set S of k columns such that

$$\|(A_S)^{-1}\| \le \frac{\sqrt{m}}{\sqrt{R}-\sqrt{k}} \left(\frac{1}{R} \sum_{i=1}^{R} \frac{1}{\sigma_i(A)^2} \right)^{1/2}. \tag{1}$$

The source proof is existential and explicitly includes this theorem in its algorithmic question.

Later algorithmic theorem

Xie's *A note on restricted invertibility with weighted columns* (arXiv:2005.01070) proves in Theorem 1.1 that for any $k \le r \le R$ and any invertible diagonal column-weight matrix W, a set S of size k satisfies

$$\mathsf{s}_{\min}(A_S W_S)^2 \ge \frac{(\sqrt{r}-\sqrt{k-1})^2}{\|W^{-1}\|_F^2} \frac{r}{\sum_{i=1}^{r} \sigma_i(A)^{-2}}.$$

The paragraph following that theorem states that its proof provides a deterministic algorithm with running time $O(k(m-k/2)n^{\theta+1})$, where θ is the matrix-multiplication exponent.

Set $W = I_m$ and $r = R$. Since $\|I_m\|_F^2 = m$, Corollary 1.2 of the later paper becomes

$$\mathsf{s}_{\min}(A_S)^2 \ge \frac{(\sqrt{R}-\sqrt{k-1})^2}{m} \frac{R}{\sum_{i=1}^{R} \sigma_i(A)^{-2}}.$$

Equivalently,

$$\|(A_S)^{-1}\| \le \frac{\sqrt{m}}{\sqrt{R}-\sqrt{k-1}} \left(\frac{1}{R} \sum_{i=1}^{R} \frac{1}{\sigma_i(A)^2} \right)^{1/2}. \tag{2}$$

Because $\sqrt{k-1} < \sqrt{k}$, estimate (2) is slightly stronger than (1). The later deterministic algorithm therefore gives a polynomial-time construction with all guarantees of the source's Theorem 11.

Scope

This is a full affirmative algorithmic answer for one of the four listed theorems, not for the entire compound question. Xie's displayed estimate does not reproduce the tail-singular-value guarantee of source Theorem 6 or the max dual-column-distance guarantee of source Theorem 9. The separate question of combining an RMS dual-column parameter with $1/\sqrt{\varepsilon}$ dependence also remains outside this identification.

References

[1] A. Naor and P. Youssef, *Restricted invertibility revisited*, arXiv:1601.00948, Theorem 11 and the algorithmic question on PDF page 8.

[2] J. Xie, *A note on restricted invertibility with weighted columns*, arXiv:2005.01070, Theorem 1.1, Corollary 1.2, and Section 4.